PAPER_712: Pillars of Creation M16 Variant 2: Post-Supernova Shock UQFF

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Abstract

A variant analysis of the Pillars of Creation incorporating the predicted supernova scenario (estimated to have detonated ~ 8000 yr ago, with the shockwave expected to reach us in ~ 1000 yr). This analysis also includes the protostellar jets at $\sim 450,000$ mph (2×10^5 m/s) that are observed in the pillar tips.

1. System Parameters

Parameter	Value	Description
M_0	$10100 M_\odot$	Initial pillar mass
r_{pillar}	4.731×10^{16} m	Pillar height
E_0^{shock}	0.15	Enhanced shock erosion
τ_{shock}	1.893×10^{14} s	Shock dissipation (6000 yr)
v_{jet}	2×10^5 m/s	Protostar jet velocity
L_{jet}	10^{28} W	Jet luminosity

2. Master UQFF Gravity Equation (v2)

$$g_{Pv2}(t) = \frac{GM(t)}{r^2}(1+Ht)\left(1-\frac{B}{B_{crit}}\right)(1-E_{shock}(t))+(U_{g1}+U_{g4})(1+f_{TRZ})+\frac{\Lambda c^2}{3}+\frac{L_{jet}}{cM_{tot}}+q(v_{je}$$

2.1 Supernova Shockwave Erosion

$$E_{shock}(t) = 0.15\, e^{-t/\tau_{shock}}$$

Stronger than the steady UV erosion ($E_0 = 0.10$), reflecting the impulsive shockwave.

2.2 Protostellar Jet Contribution

$$a_{jet} = L_{jet}/(cM_{total}) \approx 10^{-15}\,\text{m/s}^2$$

References

- Hester et al. (1996); Flagey et al. (2011), ApJ, 737, 91
- UQFF Framework, Star-Magic Session 175

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